

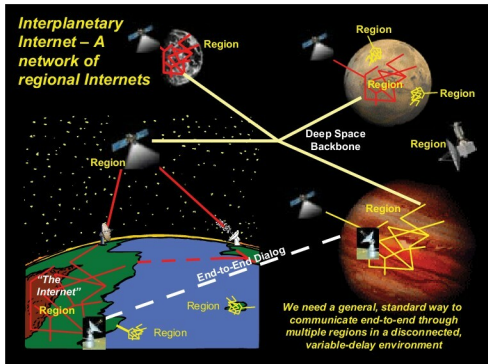
IPN: ANYWHERE AND EVERYWHERE INTERNET

The Interplanetary Internet (IPN) will be a network of Internets located on planets and moons, or aboard spacecrafts. However, several issues need to be resolved before it becomes a reality!

■ DR C.T. BHUNIA

In data communication, “anywhere and everywhere IP” is the next-generation Internet technology. The technical name of “anywhere and everywhere IP” is Interplanetary Internet (IPN). IPN is next to Internet protocol version 6 (IPv6), Internet protocol for television (IPTV), voice over IP (VoIP), compressed header IP, mobile IP and IP next generation (IPng). It is, in fact, an attempt towards a far future integration of space and terrestrial communication to permit interoperation of the Internet resident on the earth with remotely located Internets on other planets or spacecrafts in transit. Technically, IPN promises to provide Internet-like services across interplanetary distances in support of deep space exploration.

The project on IPN started way back in 1998 when US Defense Advanced Research Project Agency, as a part of ‘Next Generation Internet,’ sponsored the concept of interplanetary Internet. The supported project was seen as the future of outer space communication. Vint Cerf, who co-invented transmission control protocol/Internet protocol (TCP/IP) of the Internet technology in the recent past, said that, “It took 20 years for the Internet to take off here on the earth. It’s my guess that in next 20 years, we will want to interact with systems and people visiting the moon, mars and possibly other celestial bodies.” This may sound like science fiction, but it is becoming a fact. This article



reviews the current state of IPN.

Terrestrial versus celestial network

IPN aims to provide Internet-like services over the entire solar system (Fig. 1). It will look like the Internet on the earth and a similar one on Mars linked by gateways. These gateways will be nothing but satellites relaying the information between the Internets. How? Like terrestrial or classical Internet, IPN shall support best-effort delivery of packets over stateless infrastructure, unique global addressing with dynamic routing, end-to-end reliable service and flow control, and scalable global domain names decoupled from addressing.

On the other hand, IPN will not be Internet-like because of the differ-

ences in communication parameters between IPN and classical Internet (refer Table). The IPN protocol cannot rely on end-to-end connectivity, as in IPN end-to-end delay is huge and the cost of transmission is very high. This implies that IPN should be designed as a “network of regional Internets” spanning dissimilar environments and as a delay-tolerant network.

The Internet is a network of connected sub-networks (Fig. 2), whereas IPN is network of regional Internets (Fig. 3) where a region is an area where communication characteristics are homogeneous. In IPN, packets need to be transferred from end to end through disconnected multiple regions that tolerate variable delay. A protocol of IPN will take care of these requirements